Does Game Review on his phone agree with the routine?

The phone's, the desktop's and the routine's reviews of the same 60 games

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Abstract

On 60 of his games, Game Review on a phone or a desktop does not name the same serious errors as the routine's review of the same game. By the rule committed before any comparison, both devices differ from the routine: the phone misses 9.1% of the routine's mistakes and blunders and names 7.7% of its own that the routine does not; the desktop misses 7.3% and adds 8.7%. The next step the rule named, a better choice of which moves get the deeper second look, would not close the gap, because the devices looked twice at nearly every move they disagree on. In checks added after the results were read, nearly every disagreement is a mistake on one side and an inaccuracy on the other, and the second look itself does not repeat: on the 880 moves both the phone and the routine looked at twice, with the same nodes from the same position, the two looks gave the same numbers on 17. The phone and the desktop disagree with each other as much as either does with the routine. What tells the reviews apart is what the engine searched earlier in the game, not the phone's smaller budget. A fact about the code makes the difference last: Game Review keeps the review a device made of a game, and opens the routine's only when it has kept none. Opening the routine's once there is one would end the difference for every game the routine reviews; a second look that starts with the engine's memory cleared is the next round.

1 The question

The app reviews a game in two places. On a device, Game Review sweeps each position at 100,000 nodes on a phone or 250,000 on a desktop (a device counts as a desktop at 500 thousand nodes a second or more), gives the deeper second look to at most 25% of the game's moves, and reads no tablebase. The daily routine sweeps at 1 million nodes, gives every move it flags the second look and reads the Lichess tablebase. The second look itself is 750,000 nodes everywhere. When he opens a game on his phone, does Game Review name the same mistakes and blunders as the routine's review of it? Where it does not, is the move one the device never looked at twice?

profile	sweep (nodes)	second look	tablebase
phone	100,000	at most 25% of moves, 750,000 nodes	no
desktop	250,000	at most 25% of moves, 750,000 nodes	no
routine	1 million	every flagged move, 750,000 nodes	yes

Table 1: The three review profiles, as the app at commit `d8b771238c` sets them.

2 The data

The 60 games of the research records' frozen sample, drawn from his archive by length and result. Each was reviewed by the app's own entry point (commit `d8b771238c`, stockfish 18.0.8 stockfish-18-lite-single) at the phone, desktop and routine profiles, with a fresh engine and no cache for every game, so a document depends only on the code and the game; on one thread at fixed nodes the engine is deterministic.

3 The method

All the measures come from the review documents, with the app's own definitions. A ply is *serious* when the metrics service's `isSerious` says so of its class: a mistake, a blunder or a miss. A serious error the routine names is *missed* by a device when the device's record of the same ply is not serious; a device's serious error is *extra* when the routine's record of that ply is not. The *miss share* is the missed plies over the routine's serious errors, the *extra share* the extra plies over the device's. Beside them: the same shares for blunders and for inaccuracies, the plies whose class differs in any way, and each side's accuracy on the device against the routine. For every missed or extra ply, the device either gave it the second look or left it at the sweep's reading, and the routine's reading either came from the tablebase or did not.

Shares are pooled over the 60 games, with 95% intervals from 2,000 bootstrap draws over games (a Poisson(1) weight for each game, seed 20261014).

The rule, committed before any comparison was made, for each device: the device *agrees* with the routine when the upper ends of both the miss share's and the extra share's intervals lie below 5%; it *differs* when the lower end of either lies above 5%; otherwise the games cannot tell (*unclear*). One in twenty is the line: a phone that missed more would, every few games, leave out an error the daily review names, which a player reading both would notice within a week. When a device differs and most of its missed or extra plies are ones it left at the sweep's reading, the next round chooses better which moves get the second look, at the device's fixed budget.

Four descriptions were added after the results were read, and each is marked where it appears: whether the two profiles' second looks of the same move give the same numbers, how far apart their losses are, how near the mistake line the disagreements lie, and the phone against the desktop. They test nothing and carry no rule.

4 Results

4.1 The verdict

By the rule, both devices differ from the routine (Table 2). The phone misses 31 of the routine's 341 serious errors, 9.1% (5.9 to 12.6%), and names 26 that the routine does not, 7.7% of its own (4.9 to 10.7%). The desktop misses 7.3% (4.4 to 10.5%) and adds 8.7% (5.7 to 11.7%). The phone's miss share and the desktop's extra share each lie wholly above the 5% line. Blunders alone part about as often, and about a third of the inaccuracies differ (Table 3): on the phone, 26.1% of all moves get a different class from the routine's. Both devices read a side's accuracy a little lower than the routine does, by 0.6 and 0.7 points on average.

	serious errors						
profile	routine	device	missed	extra	miss share	extra share	verdict
Phone	341	336	31	26	9.1% (5.9, 12.6)	7.7% (4.9, 10.7)	differs
Desktop	341	346	25	30	7.3% (4.4, 10.5)	8.7% (5.7, 11.7)	differs

Table 2: The rule: serious errors (mistakes, blunders and misses) named by the routine and by each device over the 60 games, the routine's that the device missed, the device's that the routine does not name, the two shares with their 95% intervals, and the verdict against the 5% line.

	blunders		inaccuracies		any class
profile	missed	extra	missed	extra	differs
Phone	8.1% (3.7, 12.3)	6.0% (2.2, 10.0)	35.8% (29.4, 42.5)	27.0% (21.9, 32.0)	26.1% (24.0, 28.2)
Desktop	6.6% (2.7, 10.7)	6.6% (2.8, 10.8)	32.7% (27.0, 38.8)	24.6% (19.2, 29.9)	23.1% (21.1, 25.2)

Table 3: Beside the rule: the same shares for blunders alone and for inaccuracies, and the share of all moves whose class differs in any way, with 95% intervals.

4.2 Where the differences come from

Not from the second look's quota. The phone gave the second look to 26 of the 31 serious errors it missed and to all 26 it added; the desktop to 23 of 25 and all 30 (Table 4). The tablebase played almost no part. So the next round the rule named, choosing better which moves get the second look, would leave nearly every difference where it is.

	missed			extra		
profile	second look	sweep only	routine: tablebase	second look	sweep only	routine: tablebase
Phone	26	5	0	26	0	1
Desktop	23	2	0	30	0	0

Table 4: Where each missed or extra serious error comes from: whether the device gave the move the second look or left it at the sweep's reading, and whether the routine's reading came from the tablebase.

4.3 Why the reviews part

The second look is the same search on every profile: 750,000 nodes and the same candidates, from the same position with the game's history. But the engine keeps what it learned from the game's earlier searches, and each profile has searched different things before it reaches a move: the sweep at its own budget, and the second looks it chose before. In a check added after the results were read, on the 880 moves both the phone and the routine looked at twice, the two looks gave the same numbers on only 17 and reached the same depth on 260 (Table 5). Their losses for the move differ by 0.6 points at the median and 2.6 at the 90th percentile.

That is enough to move a class at the line. Of the 57 serious errors the phone and the routine disagree on, 50 are a mistake on one side and an inaccuracy on the other, and 40 have both losses within 3 points of the 10-point mistake line; on 26, one profile's own extra look, at twice the nodes, changed its verdict, so the engine had not settled them either. The desktop's are alike. And the phone and the desktop, whose profiles differ only in the sweep, disagree with each other as much as

either does with the routine: the phone misses 9.8% of the desktop's serious errors (6.5 to 13.4%) and adds 7.1% (4.6 to 9.9%).

	moves both looked at twice				serious errors they disagree on			
profile	moves	same depth	same numbers	loss gap median (90%)	moves	mistake /inaccuracy	both within 3 points	an extra look changed it
Phone	880	260	17	0.57 (2.6)	57	50	40	26
Desktop	905	245	18	0.63 (2.4)	55	54	49	25

Table 5: Why the reviews part (added after the results were read; it tests nothing). On the moves both the device and the routine looked at twice: how many, on how many the two looks reached the same depth or gave the same numbers (the best move's score and the played move's), and the gap between their losses. On the serious errors they disagree on: how many, how many are a mistake on one side and an inaccuracy on the other, how many have both losses within 3 points of the 10-point line, and on how many either profile's extra look changed its verdict.

4.4 An example

The phone's costliest miss is Black's 28... ♕d8 in Figure 1. The routine calls it a blunder (a loss of 22.3 points) and the phone an inaccuracy (6.2). Both looked at it twice, and both looks found 28... ♖c4 best at depth 16. They part on the move played. The phone's look kept 28... ♕d8 among its three candidates and scored it +1.68 there; the routine's left it out and searched it on its own, at +3.62 (both from White's side, in pawns). The routine's line is 28... ♖c4 29. ♕xc4 dxc4 30. a7 (`d4c4 e2c4 d5c4 a6a7`), from the position `6k1/5ppp/P3p1b1/3p2q1/3rP3/5P2/4Q1PP/R5K1 b - - 0 28`.

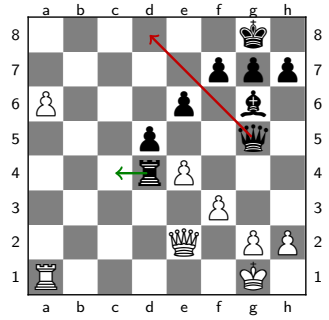


Figure 1: Black to move: 28... ♕d8 (red) where the routine's second look prefers 28... ♖c4 (green).

5 Facts about the code

These are facts about the app at commit `d8b771238c`, not measurements.

Which review Game Review opens. When he opens a game, Game Review first reads the review it kept on the device (`src/features/gamereview/GameReview.tsx`, line 203) and asks for the coach's own review only when it has kept none (line 214). The daily routine reviews the evening's games the next morning. A game he opens on his phone before then is reviewed on the phone, and

the phone keeps that review: once the routine has reviewed the game, the daily review names the routine's serious errors and Game Review still shows the phone's. For those games the differences measured here stay. A game he first opens after the routine reviewed it shows the routine's review.

The device's engine. On a device the engine runs with a 64 MB hash and keeps an evaluation cache (`src/adapters/browser/engine/index.ts`); the routine's runs with a 128 MB hash. This study ran every profile on the routine's engine with no cache, so here the phone and the desktop differ from the routine only by their profile.

The live coach's accept window. The Play screen's live coach decides whether the move played was among the fine ones with its own 3 (`src/services/livecoach/verdict.ts`, line 314), where the review imports `ACCEPT_WINDOW_WP`, also 3. The two agree today, but a change to the review's window would not reach the live coach.

6 What it means for the app

1. **The phone's review and the routine's part on about one serious error in eleven.** Nearly all of them are moves near the line between a mistake and an inaccuracy. A player reading both would meet 57 such moves in 60 games.
2. **Open the coach's review in Game Review once there is one,** in place of a review the device kept, so that his phone and the daily review name the same serious errors for every game the routine has reviewed. Before the routine reaches a game, the device's review is the only one there is.
3. **A bigger device budget would not fix it.** The desktop parts from the routine as much as the phone does, and the phone and the desktop part from each other as much. The second-look quota is not the cause either.
4. **Make the second look repeat,** for the games the routine has not reached. With the engine's memory cleared before each second look, on a hash of one size, the look becomes a function of the position, the history, the nodes and the engine build, and every profile that looks at a move twice would read it the same way. The cost is that the look no longer starts from the sweep's work, so it may search less deeply for its nodes. The next round measures that change on these 60 games against this paper's rule, with its own rule committed first.
5. **The live coach imports the review's accept window** instead of typing its own 3, so that a change to one reaches the other.

7 Limits

The 60 games are his games of one stretch of the archive, sampled for an accuracy study. The routine read here is today's: the daily reviews he has read were written by the routine on the app of their day, and this compares the profiles on one commit, so a difference is the profiles', not the app's history. Which profile his phone runs depends on its measured speed, so both device profiles are reported. Every profile ran on the routine's engine with no cache; a real device's engine has a 64

MB hash and a cache, so its second looks start from a memory that differs further, which this study did not measure. The descriptions of why the reviews part came after the results were read; they say what the measured disagreements look like and test nothing.